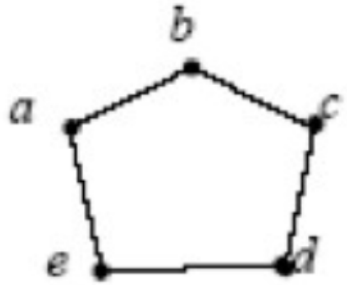
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SCSI 1013: Discrete Structure

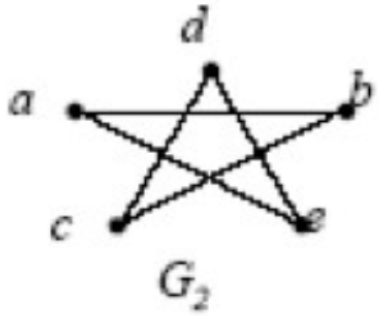
CHAPTER 4

GRAPH THEORY

Isomorphisms



G_1



G_2

- Are these two graphs (G_1 and G_2) are same?
- When we say that 2 graphs are the same mean they are isomorphic to each other.

Definition

Let $G = \{V, E\}$ and $G' = \{V', E'\}$ be graphs. G and G' are said to be isomorphic if there exist a pair of functions $f : V \to V'$ and $g : E \to E'$ such that f associates each element in V with exactly one element in V' and vice versa; g associates each element in E with exactly one element in E' and vice versa, and for each $v \in V$, and each $e \in E$, if v is an endpoint of the edge e, then $f(v)$ is an endpoint of the edge $g(e)$.

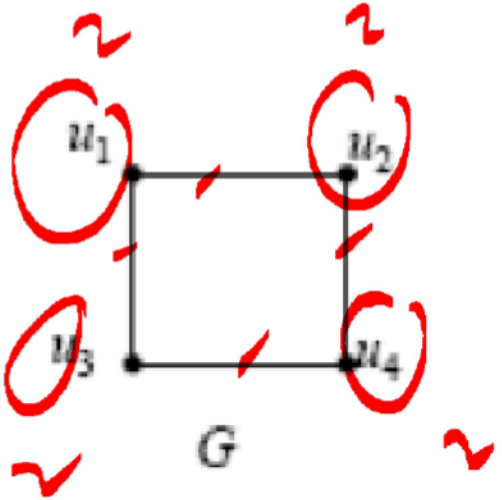
◆ If two graphs is isomorphic, they must have:

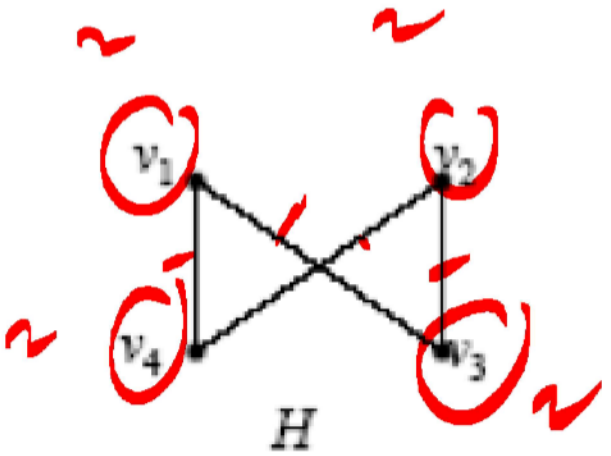
- the same number of vertices and edges,
- the same degrees for corresponding vertices,
- the same number of connected components,
- the same number of loops and parallel edges,
- both graphs are connected or both graph are not connected,
- pairs of connected vertices must have the corresponding pair of vertices connected.

◆ In general, it is easier to prove two graphs are not isomorphic by proving that one of the above properties fails.

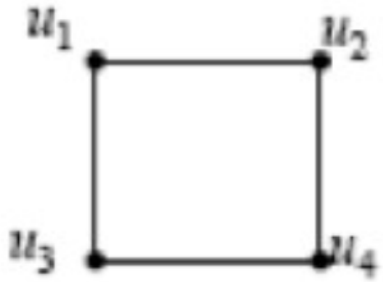
Example 1

- Determine whether G is isomorphic to H.

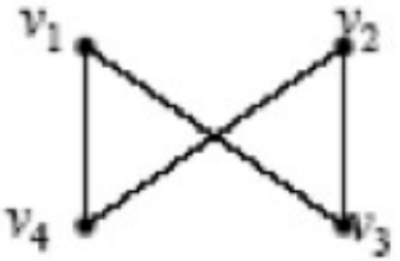




Example 1 - Solution



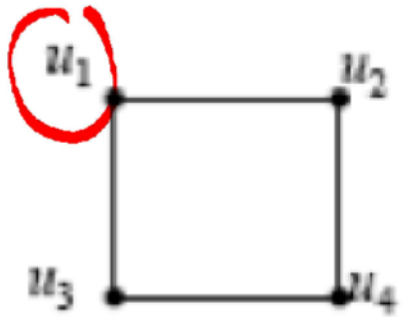
G



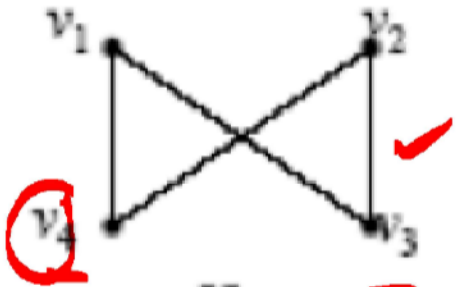
H

- Both graphs are simple and connected
- Both have 4 vertices and 4 edges
- All the vertices of both graphs have degree 2.

Example 1 - Solution



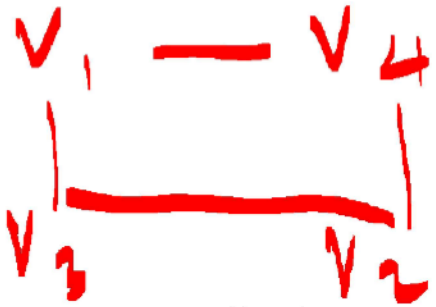
G



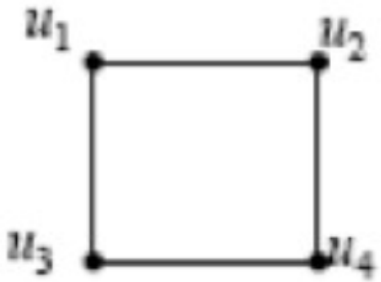
H

- Define $f : U \rightarrow V$, where $U = \{u_1, u_2, u_3, u_4\}$ and $V = \{v_1, v_2, v_3, v_4\}$;

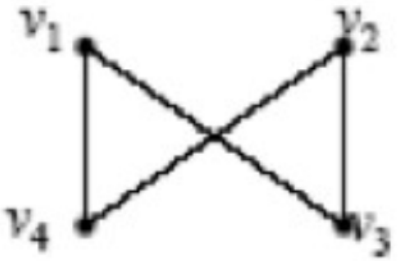




$f(u_1) = v_1 \quad f(u_2) = v_4 \quad f(u_3) = v_3 \quad f(u_4) = v_2$



G



H

- To verify whether G and H are isomorphic, we examine the adjacency matrix A_G with rows and columns labeled in the order u_1, u_2, u_3, u_4, and the adjacency matrix A_H with rows and columns labeled in the order v_1, v_4, v_3, v_2.

$$f(u_1) = v_1 ; f(u_2) = v_4 ; f(u_3) = v_3 ; f(u_4) = v_2 .$$

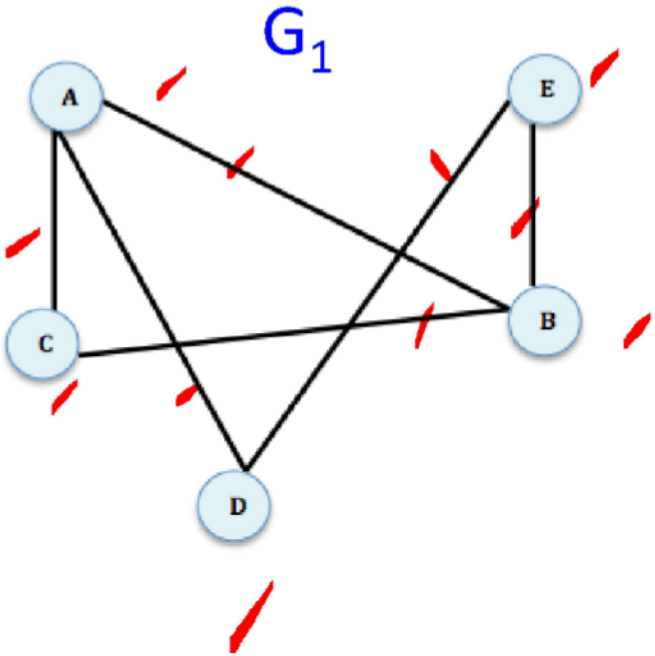
- A_G and A_H are the same, G and H are isomorphic.

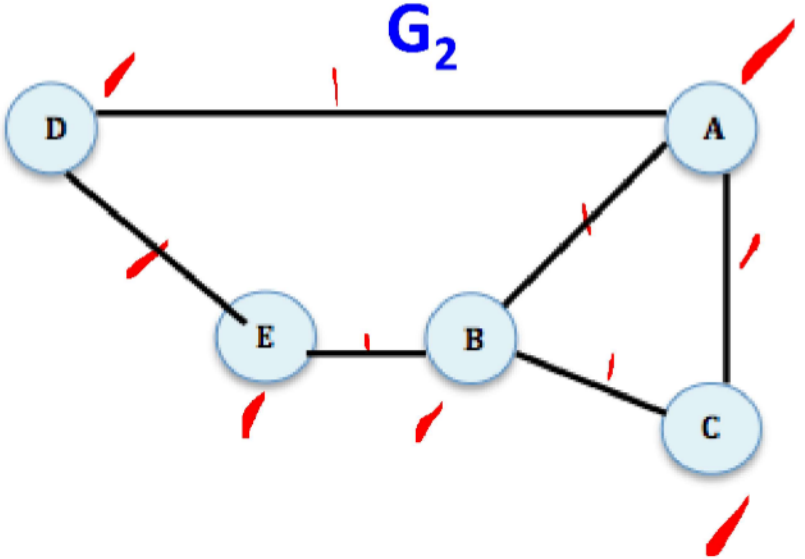
$$A_G = \begin{array}{c} \\ u_1 \\ u_2 \\ u_3 \\ u_4 \end{array} \begin{array}{c} \begin{array}{cccc} u_1 & u_2 & u_3 & u_4 \end{array} \\ \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix} \end{array} \qquad A_H = \begin{array}{c} \\ v_1 \\ v_4 \\ v_3 \\ v_2 \end{array} \begin{array}{c} \begin{array}{cccc} v_1 & v_4 & v_3 & v_2 \end{array} \\ \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix} \end{array}$$

Thus G and H are isomorphic

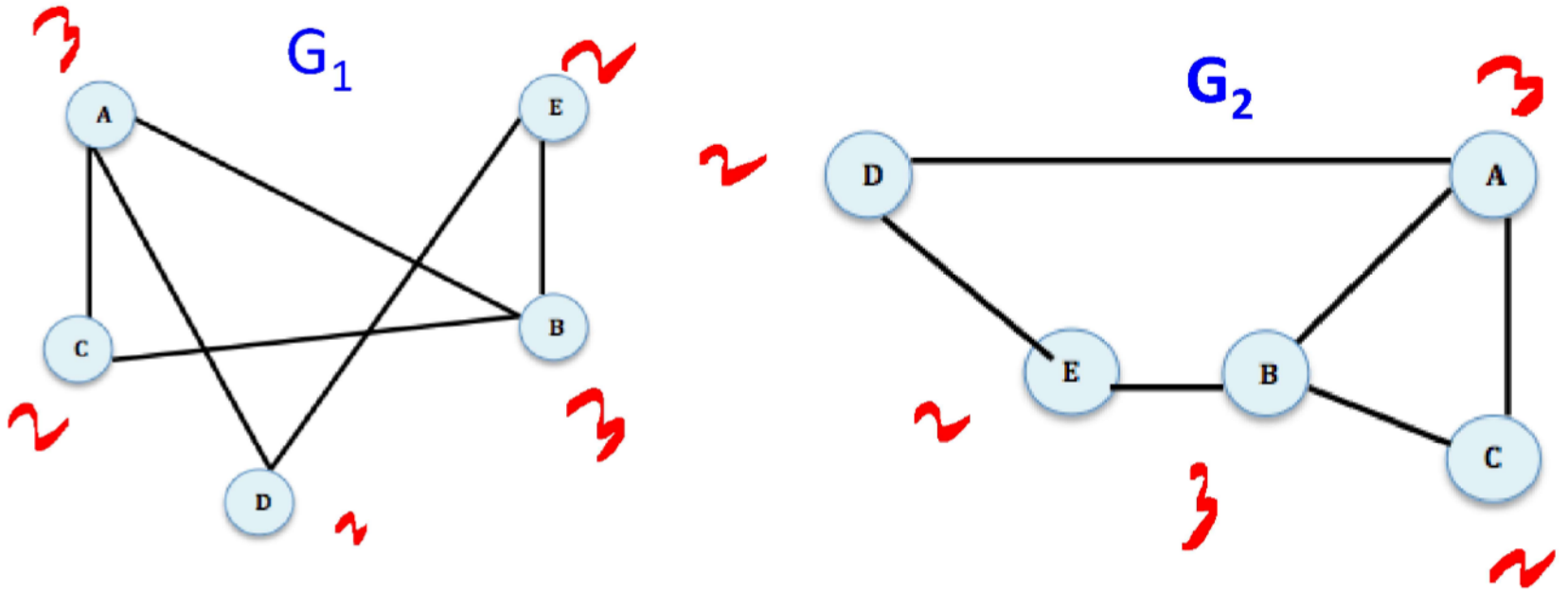
Example 2

Q: Show that the following two graphs are isomorphic.



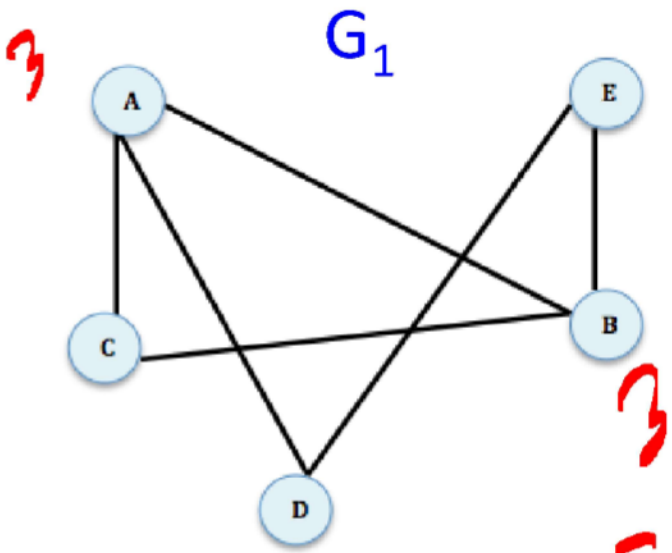


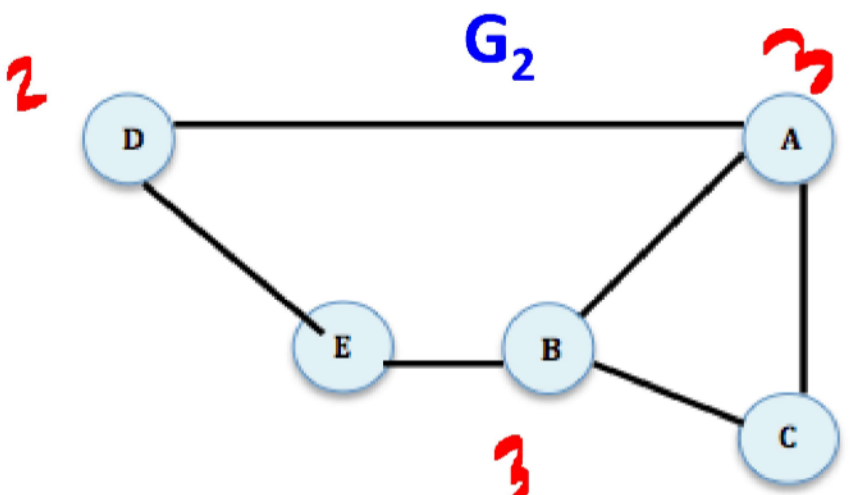
Solution

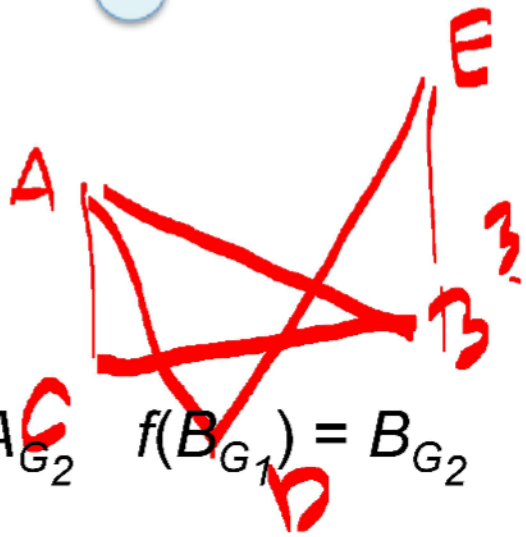


- Both have 5 vertices and 6 edges
- Both are connected and simple graph
- Both have 2 vertices with 3 degree and 3 vertices with 2 degree

Solution









$f(A_{G_1}) = A_{G_2}$ $f(B_{G_1}) = B_{G_2}$ $f(C_{G_1}) = C_{G_2}$ $f(D_{G_1}) = D_{G_2}$ $f(E_{G_1}) = E_{G_2}$

A_{G1} :

$$\begin{array}{c} \\ A \\ B \\ C \\ D \\ E \end{array}\begin{array}{c} \begin{array}{ccccc} A & B & C & D & E \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

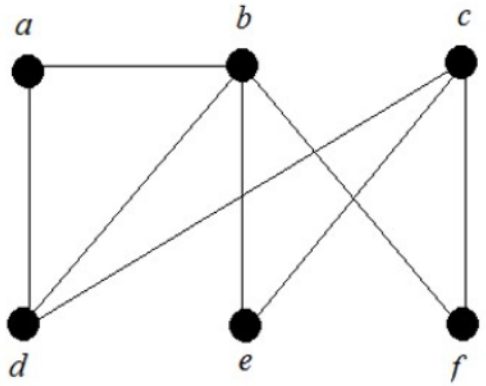
dan

$$\begin{array}{c} \\ A \\ B \\ C \\ D \\ E \end{array}\begin{array}{c} \begin{array}{ccccc} A & B & C & D & E \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

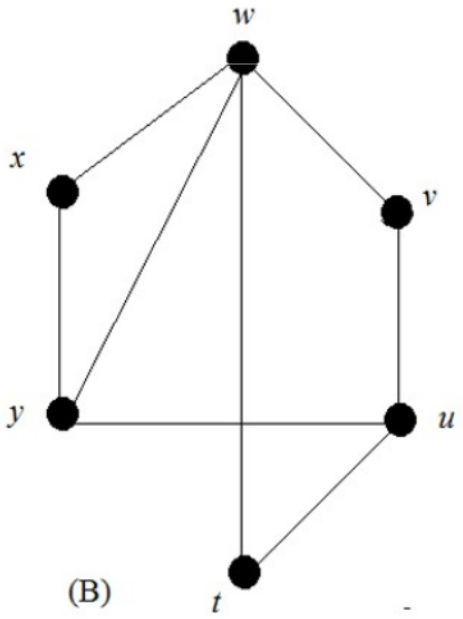
Thus, G_1 and G_2 are isomorphic

Exercise
Past Year 2015/2016

Determine whether the graphs in Figure 2 (A and B) are isomorphic. If the graphs are isomorphic, find their adjacency matrices; otherwise, give an invariant that the graphs do not share. (6 marks)



(A)



(B)